

§7.3 - Trigonometric Substitution

Example #1

If

$$\tan(t) = \frac{\sqrt{9 - x^2}}{x}$$

for $x \in (0, 3)$ what is

$$\sin(t)$$

in terms of x ?

Example #2 (Integral involving $a^2 - x^2$)

Find

$$\int \sqrt{1 - x^2} \, dx.$$

Example #3 (Integral involving $a^2 - x^2$)

Find

$$\int \sqrt{36 - x^2} \, dx.$$

Example #4 (Integral involving $a^2 + x^2$)

Find

$$\int \frac{1}{9^2 + x^2} dx.$$

Example #5 (Integral involving $a^2 + x^2$)

Find

$$\int \frac{1}{\sqrt{16 + x^2}} dx.$$

Example #6 (Integral involving $x^2 - a^2$)

Find

$$\int \frac{1}{\sqrt{x^2 - 49}} dx$$

for $x > 7$.

Example #7

Express

$$x^2 - 2x + 10$$

as

$$u^2 + h$$

where $u = x - k$

Example #8

Find

$$\int \frac{1}{x^2 - 2x + 10} dx.$$

Remark

Here is the summary of the substitutions:

1. $a^2 - x^2$: use $x = a \sin(t)$ and $\cos^2(t) + \sin^2(t) = 1$
2. $a^2 + x^2$: use $x = a \tan(t)$ and $1 + \tan^2(t) = \sec^2(t)$
3. $x^2 - a^2$: use $x = a \sec(t)$ and $1 + \tan^2(t) = \sec^2(t)$

Don't forget,

$$\int \sec(x) \, dx = \ln |\sec(x) + \tan(x)| + C.$$